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# Mobile Application Development - MCA221IA

## Project Based Laboratory

# College Forum App

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# Introduction

- At RVCE, important updates are scattered across WhatsApp groups, emails and physical boards.
- College Forum App is a Progressive Web Application that centralizes all campus notices.
  - In the user interface it is branded as “RVCE Connect”.
- Three roles: Admin, Publisher (faculty / HODs / club heads) and Viewer (student).
- Departments and clubs act as communities that students subscribe to.
- Mobile-first, installable, and works offline.



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## Key Features:

- Role-based access control – Admin, Publisher, Viewer.
- Community-targeted notices for departments and clubs.
- Subscriptions, a personalized feed and post likes.
- Real-time Web Push notifications via a per-community bell.
- Post scheduling and automatic expiry / archiving.
  - Admin-only post deletion, recorded in an audit log.
- PWA: offline access through service workers; installable.
- Secure JWT authentication with bcrypt-hashed passwords.



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# Technology Stack

- Frontend: HTML5, CSS3, Bootstrap 5, Vanilla JavaScript.
- Backend: Node.js with Express.js (RESTful APIs).
- Database: MySQL 8.
- Authentication & security: JSON Web Tokens (JWT), bcrypt.
- PWA: Web App Manifest, Service Workers, Cache Storage, Web Push (VAPID).
- Tooling: VS Code, npm, Git & GitHub, Docker (MySQL).



# Requirements & Features

- Functional requirements:
  - Authentication, registration and role-based access (JWT).
  - User and community management by the administrator.
  - Notice publishing targeted to chosen communities.
  - Personalized feed, likes and subscriptions.
  - Push notifications and offline access.
- Non-functional requirements:
  - Performance, security, scalability, usability and responsive compatibility.

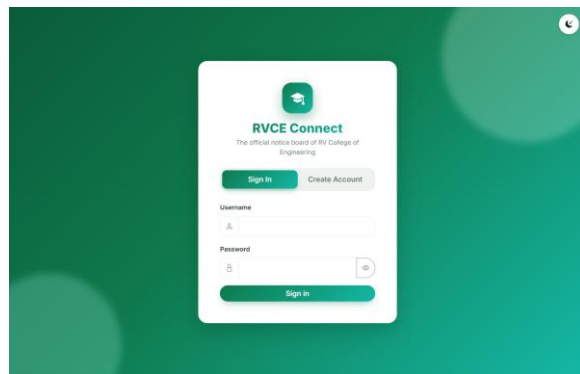


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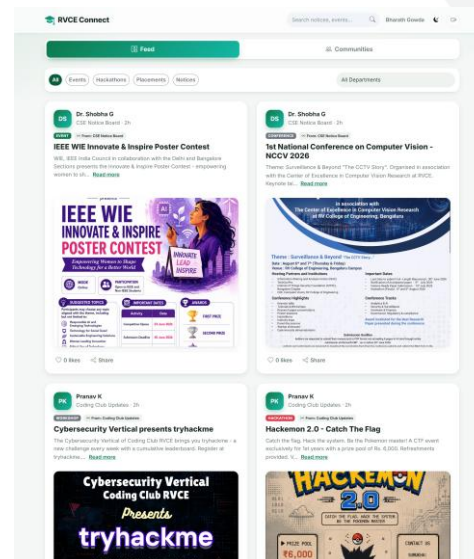
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# User Interface Mockups / Wireframes

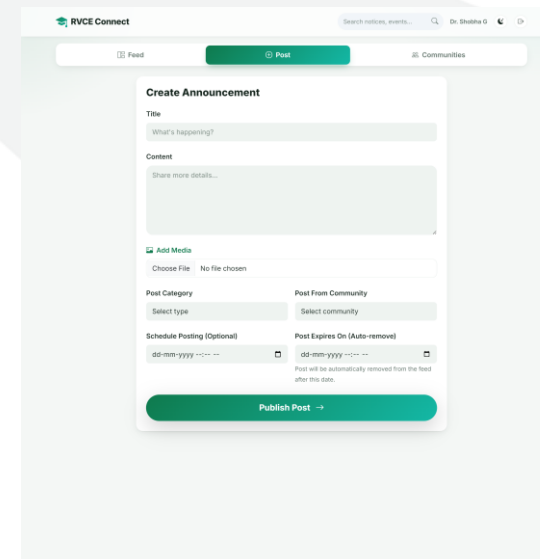
- Navigation flow:
  - Login authenticates and routes users to a role-based dashboard.
  - Viewer: Feed ↔ Communities – subscribe, like and read notices.
  - Publisher: compose announcements; Admin: dashboard and moderation.



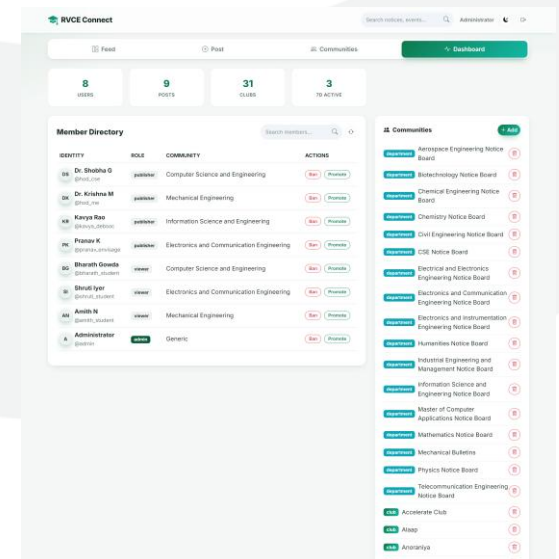
Login screen



Student feed



Publisher – compose



Admin dashboard



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# Demonstration

- Live (development): <http://localhost:3000>
- Flow: register / login → browse feed → like & subscribe.
- Publisher posts a notice → subscribers get a push notification.
- Disconnect the network → cached notices remain available (offline).

**RVCE Connect**  
The official notice board of RV College of Engineering

Sign In Create Account

Full Name Username

College Email (@rvce.edu.in)

Department  
Select a department

Password

Create Account

*Self-registration*

**RVCE Connect**

Feed Post Communities

Create Announcement

Title  
What's happening?

Content  
Share more details...

Add Media  
Choose File No file chosen

Post Category  
Select type

Post From Community  
Select community

Schedule Posting (Optional)  
88-mm-yyyy --:--

Post Expires On (Auto-removed)  
88-mm-yyyy --:--

Publish Post →

*Campus feed*





# Conclusion

- Delivered a secure, centralized campus communication platform.
- Role-based broadcasting reduces information overload.
- PWA adds offline access, push notifications and installability.
- Learning: full-stack development, database design, JWT auth, PWA and UI/UX.
- Future scope: event RSVP, analytics, advanced search, ERP/LMS integration.



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# Thank You